



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

Rows in the dataset

18

Columns

Features tracked

37

Missing Values

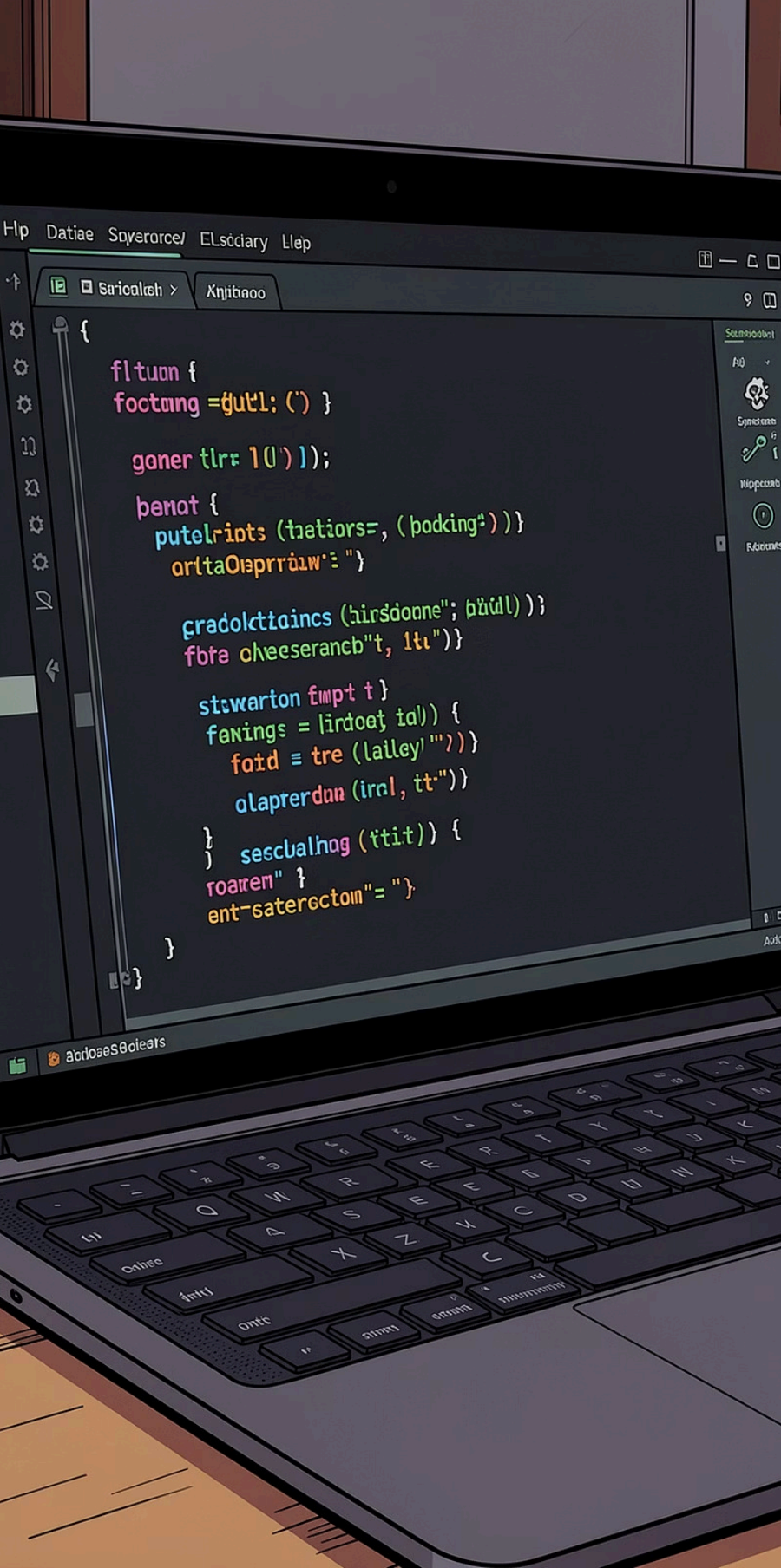
In Review Rating only

Customer Data

- Age, Gender, Location
- Subscription Status
- Previous Purchases & Frequency

Purchase Data

- Item, Category, Amount
- Season, Size, Color
- Discount, Payment Method, Shipping



Data Preparation in Python

01

Load & Explore

Imported via pandas; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean Missing Data

Imputed 37 missing Review Ratings using the median rating per product category.

03

Standardize & Engineer

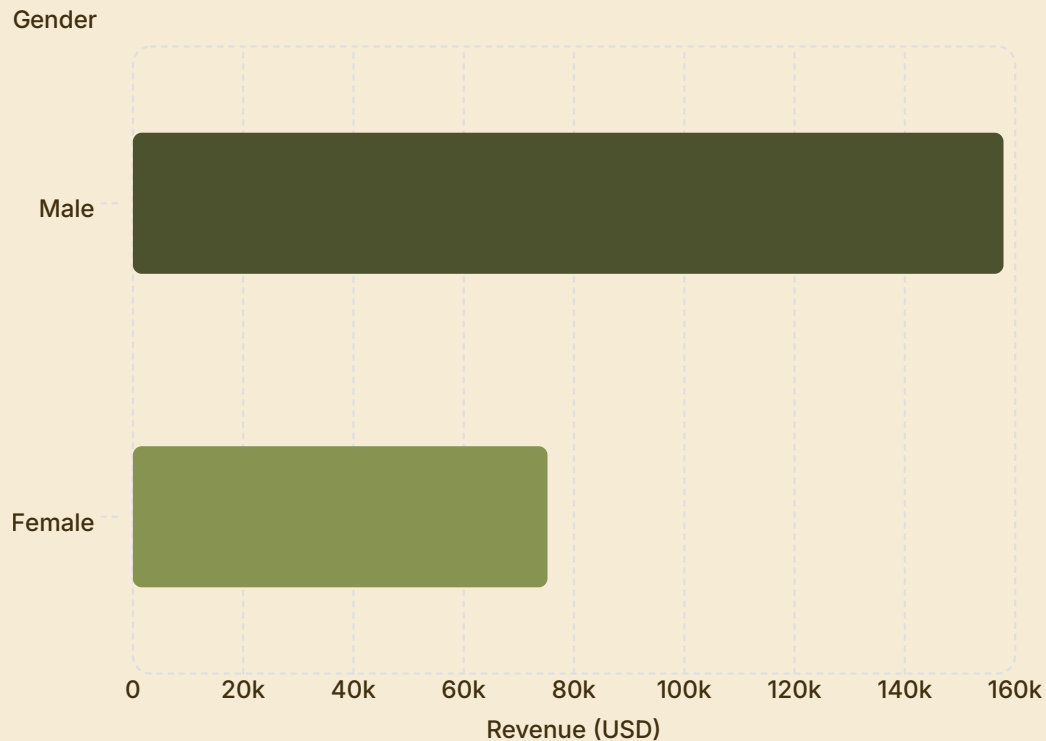
Renamed columns to snake_case; created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Revenue & Spending Insights



Revenue by Gender

Male customers generated **\$157,890** in revenue — more than double female customers at **\$75,191**.

High-Spending Discount Users

Customers who used discounts but still spent **above the average purchase amount** were identified — revealing discount-resilient high-value shoppers.

Product Ratings & Shipping Behavior



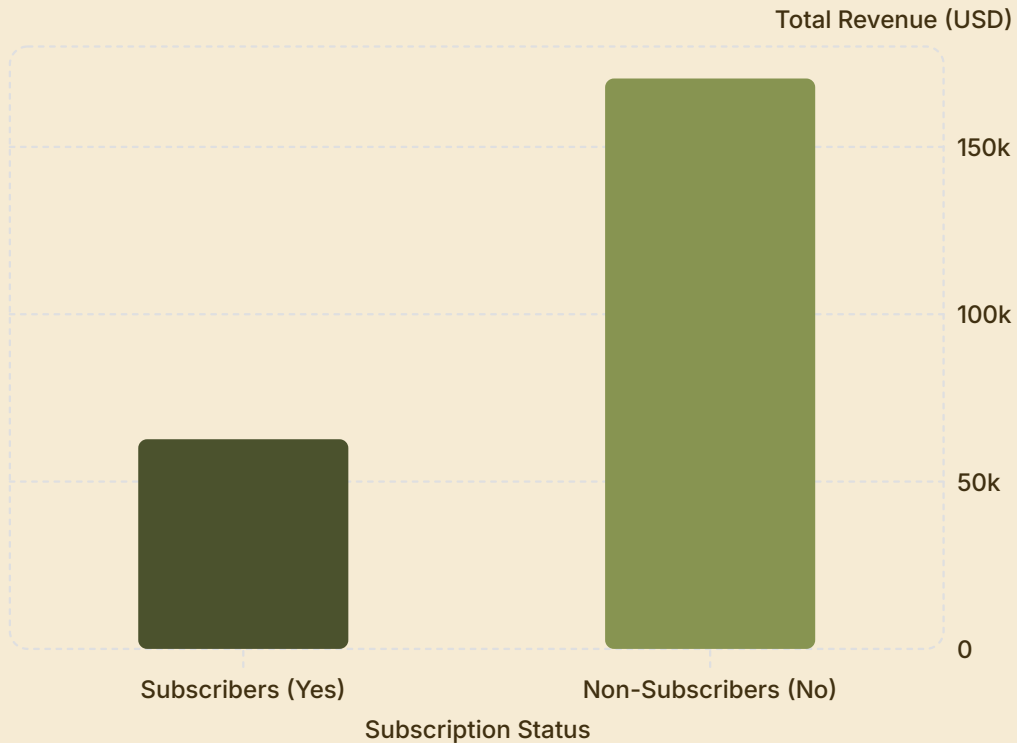
Top 5 Products by Rating

Gloves lead with a **3.86** average rating. Overall average rating across all products: **3.75**.

Shipping Type Comparison

Express shipping averages **\$60.48** per order vs. Standard at **\$58.46** — a modest but consistent premium.

Subscribers vs. Non-Subscribers



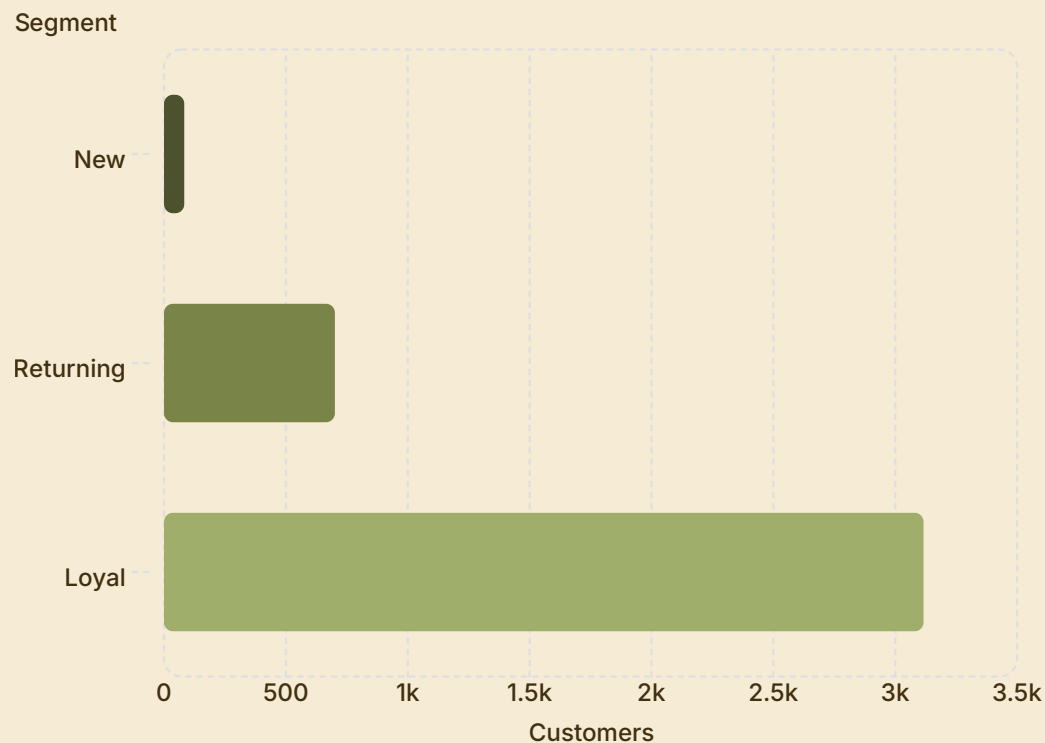
Key Findings

Non-subscribers drive **73%** of total revenue (\$170,436) despite a nearly identical average spend (\$59.87 vs. \$59.49).

Only **27%** of customers (1,053) are subscribers — a major growth opportunity.

- Of customers with more than 5 purchases, **958 are subscribers** vs. 2,518 non-subscribers — repeat buyers show higher subscription uptake.

Customer Segmentation & Discount Trends



Segment Breakdown

Loyal customers dominate at 3,116 (80%), with 701 Returning and only 83 New customers.

Most Discount-Dependent Products

Hat (50%), Coat (49%), Sneakers (49%), Sweater (48%), and Pants (47%) have the highest share of discounted purchases — indicating price sensitivity.



Top Products by Category

Accessories

1. Jewelry — 171 orders
2. Sunglasses — 161 orders
3. Belt — 161 orders

Clothing

1. Blouse — 171 orders
2. Pants — 171 orders
3. Shirt — 169 orders

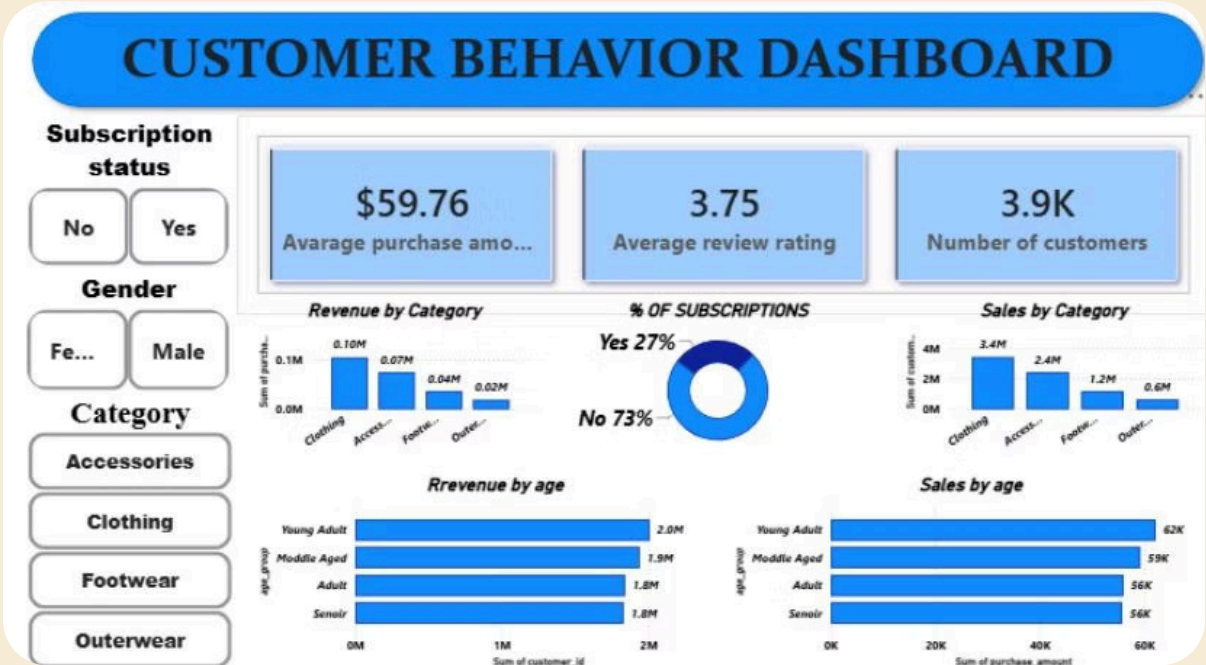
Footwear

1. Sandals — 160 orders
2. Shoes — 150 orders
3. Sneakers — 145 orders

Outerwear

1. Jacket — 163 orders
2. Coat — 161 orders

Power BI Dashboard



Interactive Customer Behavior Dashboard

Built in Power BI with filters for Subscription Status, Gender, and Category.

Visuals include Revenue by Category, Subscription %, Sales by Category, and Revenue & Sales by Age Group.

\$59.76

Average purchase amount

3.75

Average review rating

3.9K

Total customers

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to grow the 27% base.



Loyalty Programs

Reward repeat buyers to convert Returning customers into Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on discount-heavy products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.



Product Positioning

Highlight top-rated and best-selling products in campaigns.